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(54) **CATALYTIC CONVERTER ANTI-THEFT
SYSTEMS AND METHODS**

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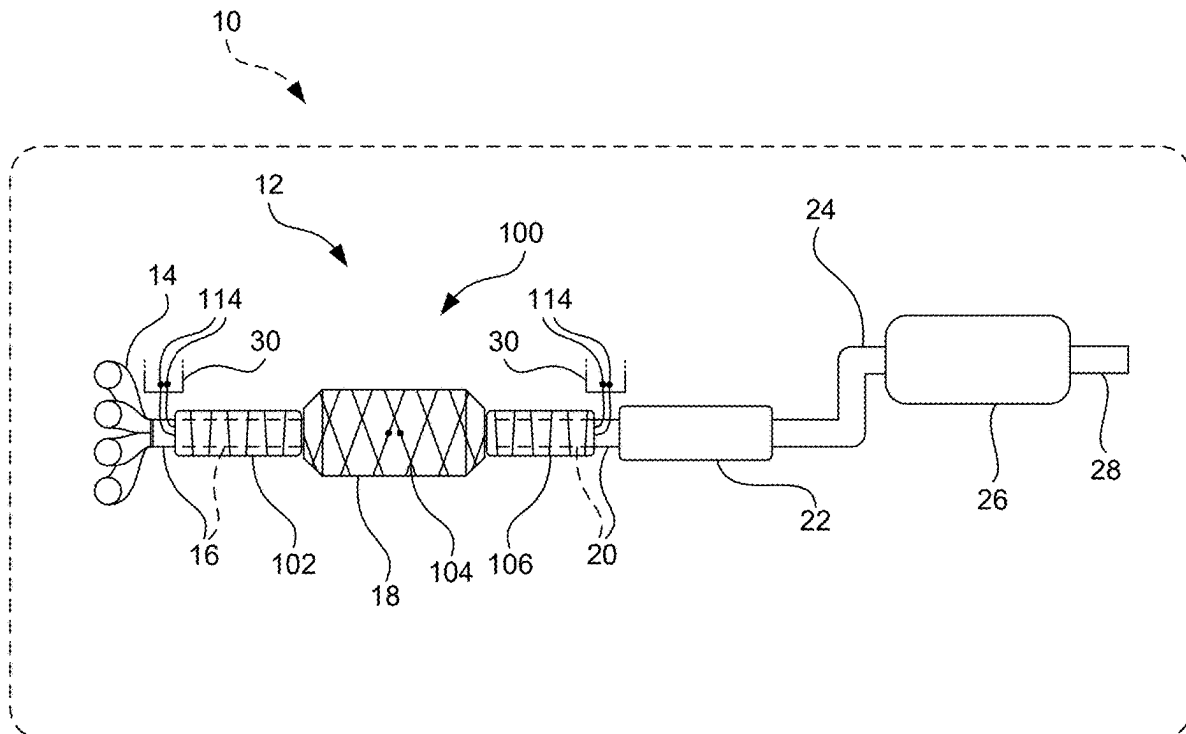
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(57) **ABSTRACT**

A catalytic converter anti-theft system for an exhaust system of a vehicle includes a thermally insulating first layer configured to be disposed about at least a portion of the exhaust system. A cable is configured to be disposed about the thermally insulating first layer. A cut and abrasion resistant second layer is configured to be disposed about the cable. An abrasion resistant third layer is configured to be disposed about the cut and abrasion resistant second layer.

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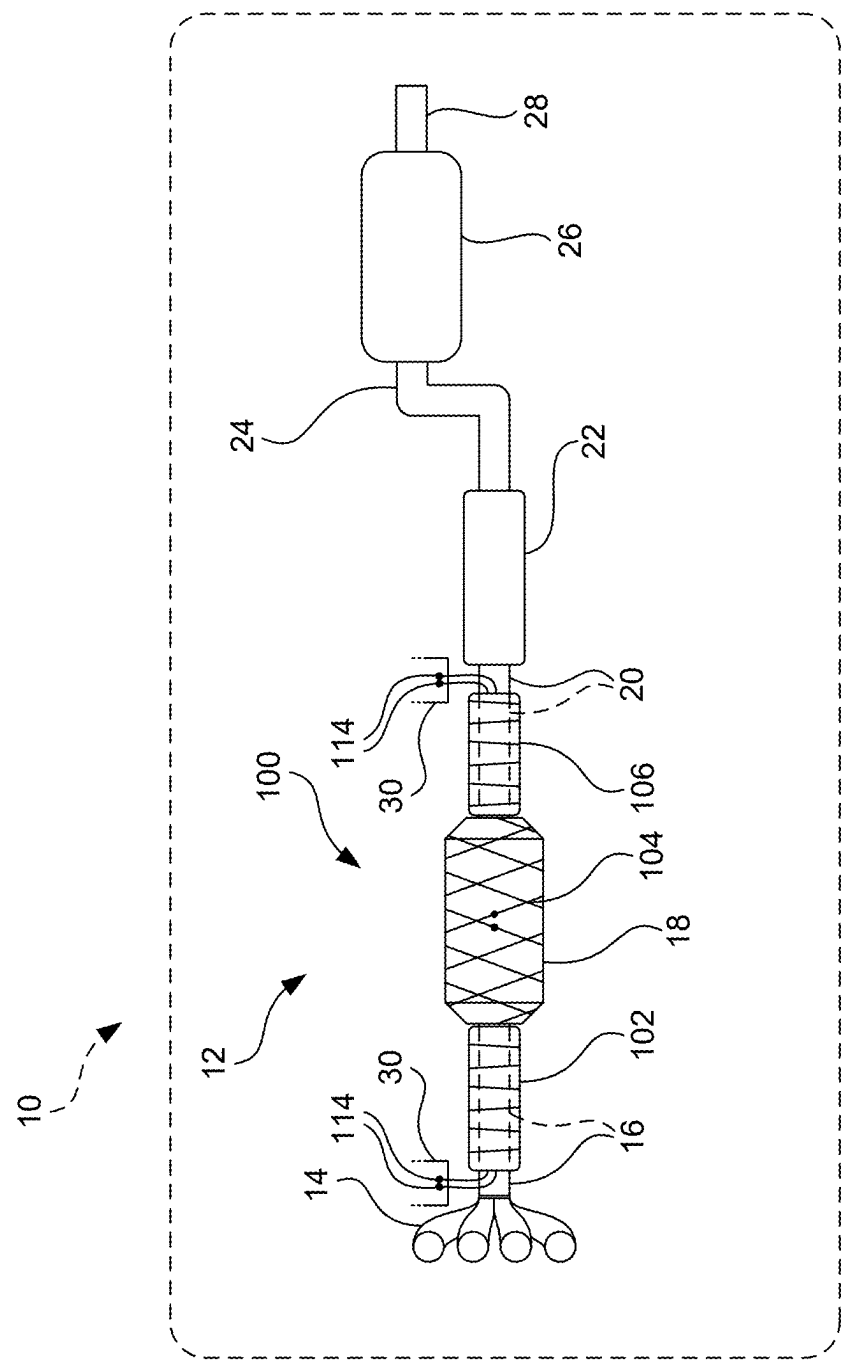


FIG. 1

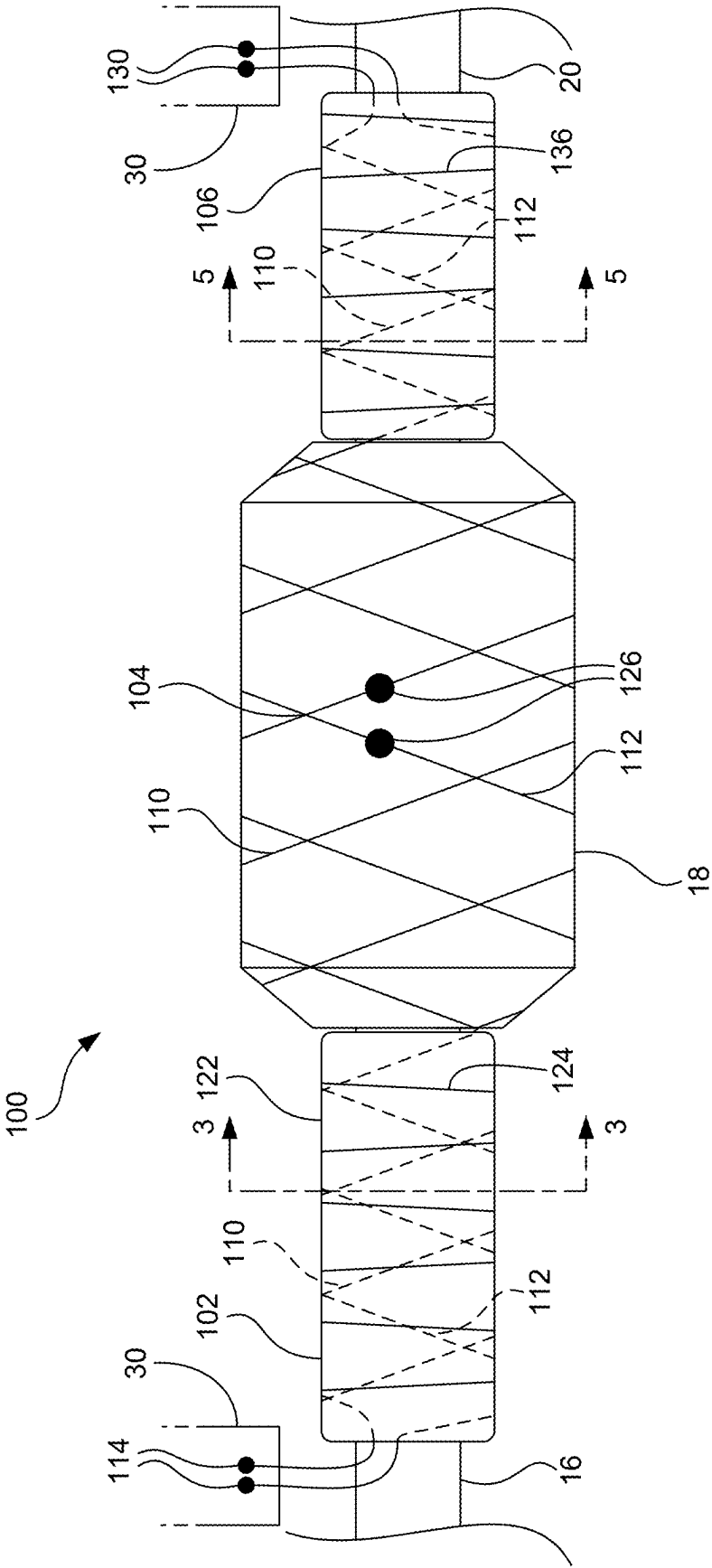


FIG. 2

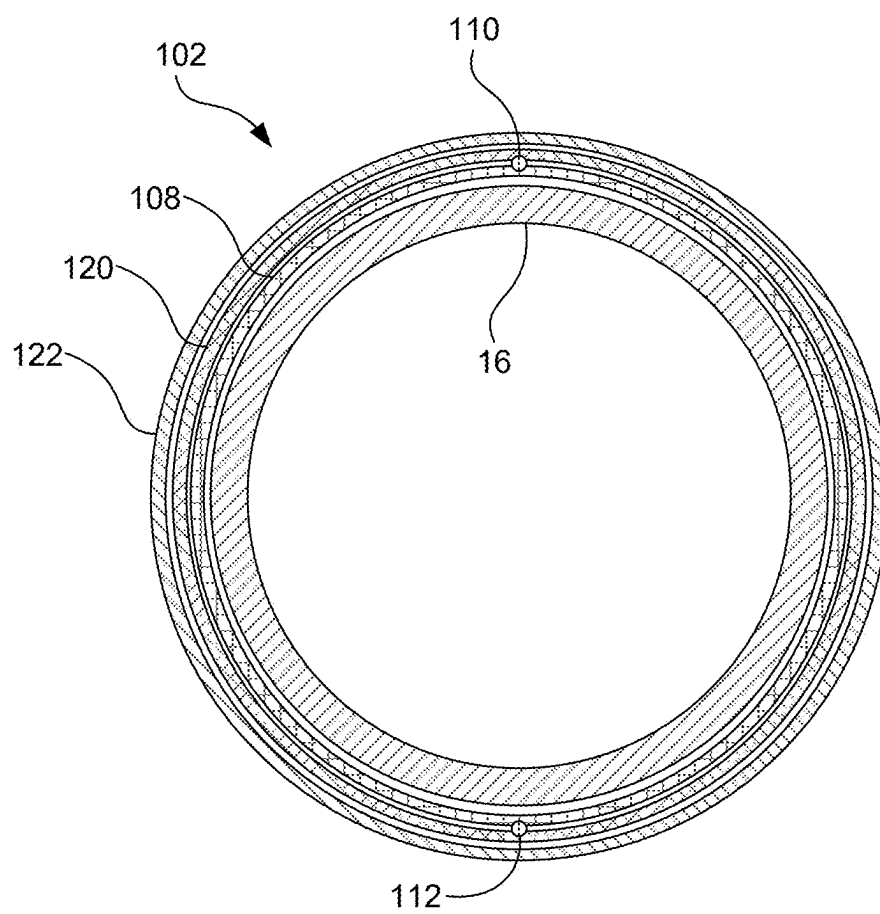


FIG. 3

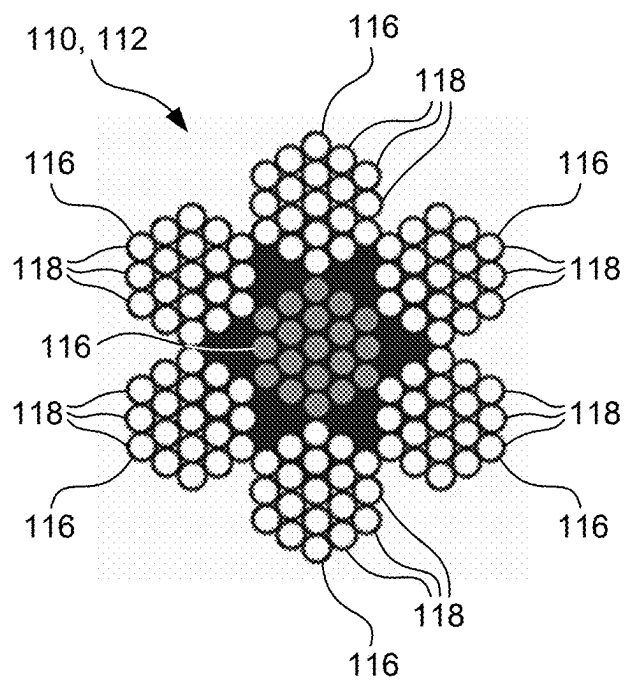


FIG. 4

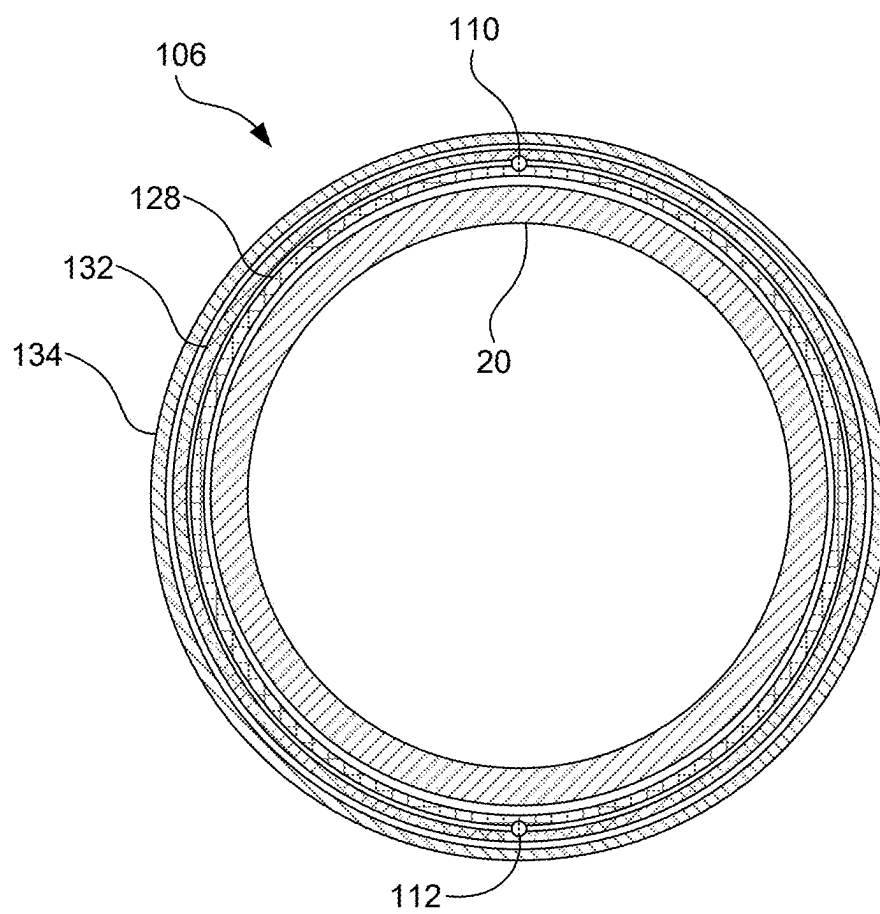


FIG. 5

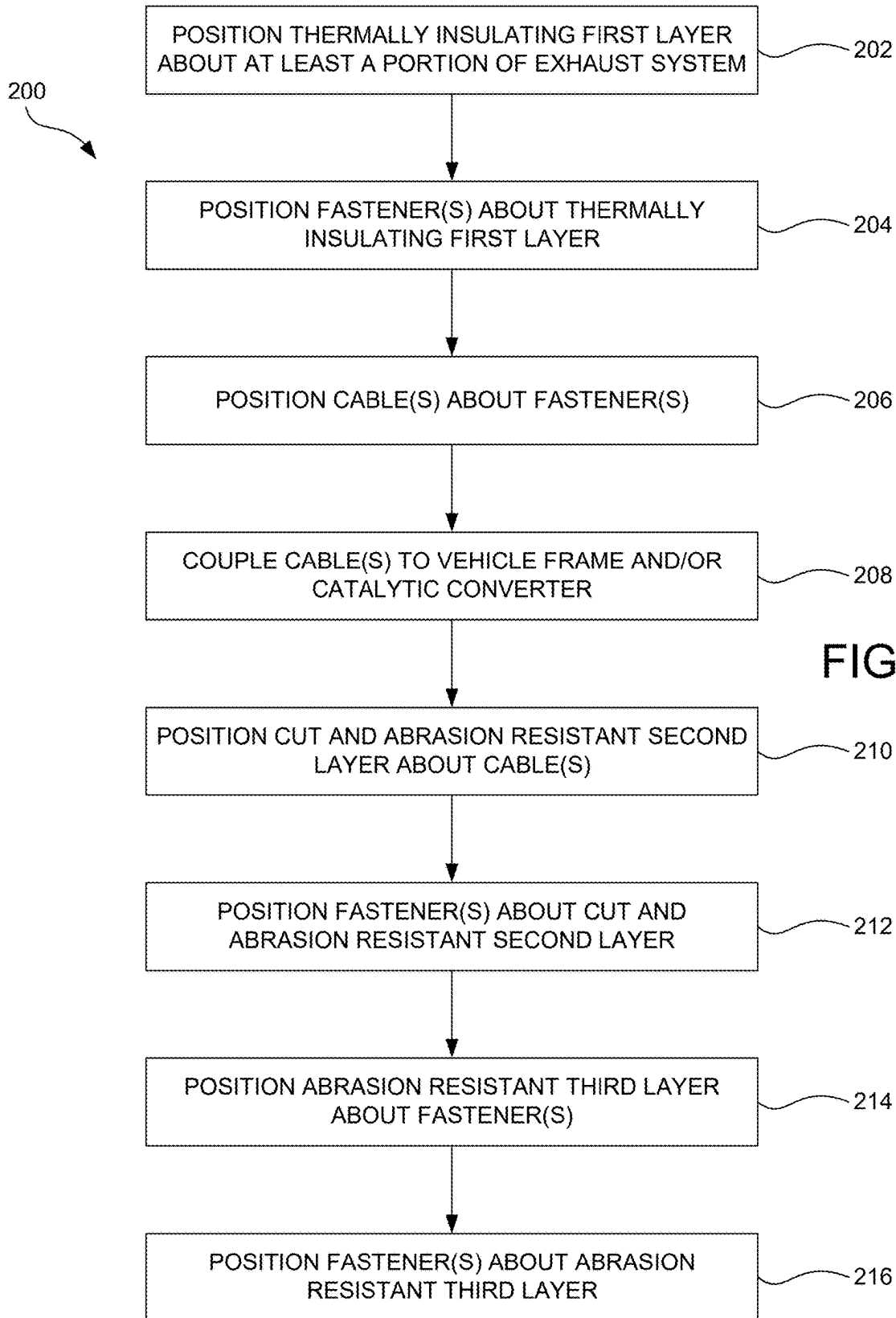


FIG. 6

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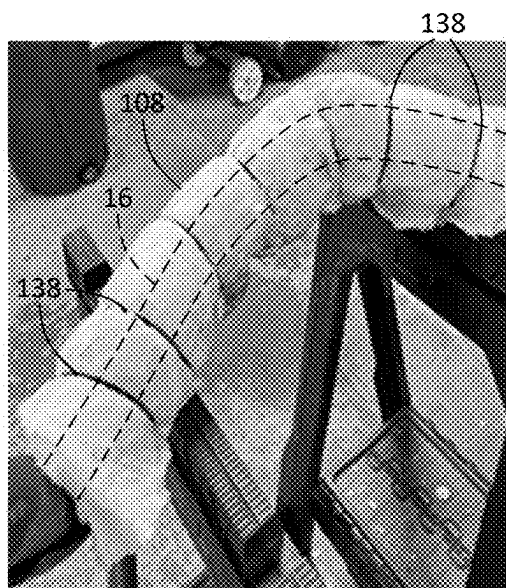


FIG. 7

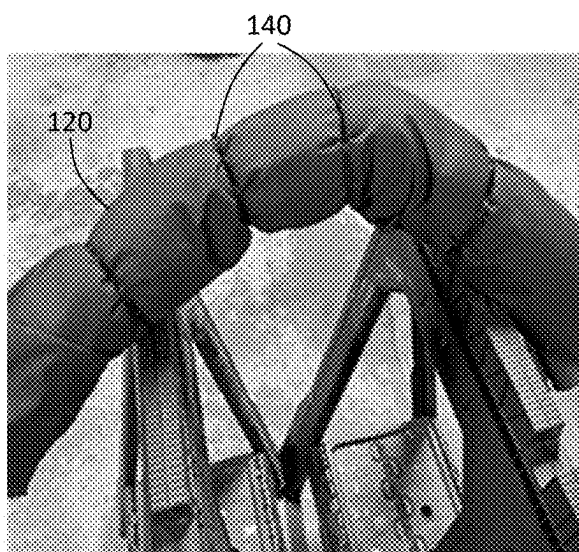


FIG. 8

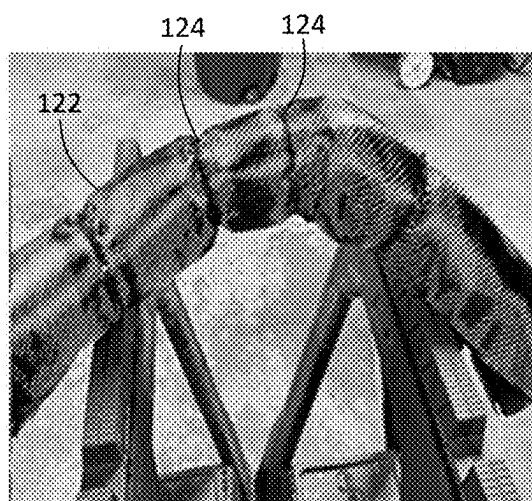


FIG. 9

CATALYTIC CONVERTER ANTI-THEFT SYSTEMS AND METHODS

BACKGROUND

[0001] Catalytic converters are commonly detached from vehicles and stolen because they are constructed from valuable metals, such as rhodium, palladium, and platinum, which can be recycled and resold for several to tens of thousands of dollars per ounce. These thefts cost the automotive industry tens of millions of dollars each year.

[0002] Thieves typically detach catalytic converters from vehicles by cutting through exhaust conduits on opposite sides of the catalytic converters. Accordingly, some previous anti-theft systems attempted to deter and inhibit thieves by at least partially surrounding the exhaust conduits and/or catalytic converters with one or more cables. However, such systems typically require significant amounts of time to install. In addition, some saws can nevertheless be maneuvered through spaces between cable sections, and the exhaust conduits may nevertheless be cut, albeit with more difficulty.

[0003] It is with respect to these and other general considerations that embodiments have been described. Also, although relatively specific problems have been discussed, it should be understood that the embodiments should not be limited to solving the specific problems identified in the background.

SUMMARY

[0004] Aspects of the present disclosure are directed to catalytic converter anti-theft systems and methods that deter and/or inhibit cutting adjacent exhaust conduits (for example, using powered saws, grinders, snips and the like) to permit detachment of catalytic converters from vehicles.

[0005] In one aspect, a catalytic converter anti-theft system, according to an example embodiment, for an exhaust system of a vehicle is disclosed. The system includes a thermally insulating first layer configured to be disposed about at least a portion of the exhaust system; a cable configured to be disposed about the thermally insulating first layer; a cut and abrasion resistant second layer configured to be disposed about the cable; and an abrasion resistant third layer configured to be disposed about the cut and abrasion resistant second layer.

[0006] In another aspect, a catalytic converter anti-theft system, according to another example embodiment, for an exhaust system of a vehicle is disclosed. The system includes a thermally insulating first layer configured to be disposed about at least a portion of the exhaust system; a cut and abrasion resistant second layer configured to be disposed about the thermally insulating first layer; an abrasion resistant third layer configured to be disposed about the cut and abrasion resistant second layer; and a plurality of fasteners configured to couple the anti-theft system to the vehicle.

[0007] In yet another aspect, a method for coupling a catalytic converter anti-theft system to an exhaust system of a vehicle, according to yet another example embodiment, is disclosed. The method includes: positioning a thermally insulating first layer about at least a portion of the exhaust system; positioning a cable about the thermally insulating first layer; coupling the cable to the vehicle; positioning a cut and abrasion resistant second layer about the cable; posi-

tioning an abrasion resistant third layer about the cut and abrasion resistant second layer; and coupling the abrasion resistant third layer to the vehicle using a plurality of fasteners.

[0008] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS/FIGURES

[0009] Non-limiting and non-exhaustive examples are described with reference to the following Figures.

[0010] FIG. 1 shows a diagrammatic view of a catalytic converter anti-theft system, according to an example embodiment, coupled to an exemplary exhaust system of an automotive vehicle.

[0011] FIG. 2 shows a detail diagrammatic view of the catalytic converter anti-theft system and the vehicle of FIG. 1.

[0012] FIG. 3 shows a cross-sectional view of the catalytic converter anti-theft system taken along line 3-3 of FIG. 2.

[0013] FIG. 4 shows a cross-sectional view of a cable of the catalytic converter anti-theft system of FIG. 1.

[0014] FIG. 5 shows a cross-sectional view of the catalytic converter anti-theft system taken along line 5-5 of FIG. 2.

[0015] FIG. 6 shows a flow diagram of a method for coupling a catalytic converter anti-theft system to an exhaust system of a vehicle, according to an example embodiment.

[0016] FIG. 7 shows a thermally insulating first layer and fasteners positioned about an exhaust conduit, according to the method of FIG. 6.

[0017] FIG. 8 shows a cut and abrasion resistant second layer and fasteners positioned about the thermally insulating first layer and fasteners of FIG. 7, according to the method of FIG. 6.

[0018] FIG. 9 shows an abrasion resistant third layer and fasteners positioned about the cut and abrasion resistant second layer and fasteners of FIG. 8, according to the method of FIG. 6.

DETAILED DESCRIPTION

[0019] In the following detailed description, references are made to the accompanying drawings that form a part hereof, and in which are shown by way of illustrations specific embodiments or examples. These aspects may be combined, other aspects may be utilized, and structural changes may be made without departing from the present disclosure. Embodiments may be practiced as methods, systems, or devices. The following detailed description is not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims and their equivalents.

[0020] FIG. 1 shows a diagrammatic view of an automotive vehicle 10, for example a commercial vehicle or a passenger vehicle, and a catalytic converter anti-theft system 100, according to an example embodiment, coupled to an exhaust system 12 of the vehicle 10. As used herein, the term “vehicle” and “automotive vehicle” broadly refer to any motor vehicle that includes an exhaust system, including cars, trucks, buses, vans, etc. Illustratively, the exhaust system 12 of the vehicle 10 includes an exhaust manifold 14

in fluid communication with and receiving discharged gases from an engine (not shown) of the vehicle 10, for example an internal combustion engine such as a diesel engine or a gasoline engine. The illustrative exhaust system 12 further includes a first or upstream exhaust conduit 16 in fluid communication with the exhaust manifold 14, a catalytic converter 18 in fluid communication with the upstream exhaust conduit 16, and a second or downstream exhaust conduit 20 in fluid communication with the catalytic converter 18. Illustratively, the exhaust system 12 may further include a resonator 22 in fluid communication with the downstream exhaust conduit 20, a third or “catback” exhaust conduit 24 in fluid communication with the resonator 22, a muffler 26 in fluid communication with the catback exhaust conduit 24, and a tailpipe 28 in fluid communication with the muffler 26 for discharging exhaust gases from the vehicle 10. In some cases, the exhaust system 12 may include additional or alternative components.

[0021] With continued reference to FIG. 1 and additional reference to FIG. 2, the catalytic converter anti-theft system 100 is disposed about at least a portion, or one or more components, of the exhaust system 12. Illustratively, the catalytic converter anti-theft system 100 includes a first section 102 that is disposed about at least a portion of the first exhaust conduit 16, a second section 104 that is disposed about the catalytic converter 18, and a third section 106 that is disposed about at least a portion of the second exhaust conduit 20. The first section 102, the second section 104, and the third section 106 of the system 100 together deter unauthorized detachment of the catalytic converter 18 from the vehicle 10 by inhibiting cutting through the first exhaust conduit 16 and the second exhaust conduit 20.

[0022] With further reference to FIG. 2 and additional reference to FIG. 3, the first section 102 of the catalytic converter anti-theft system 100 is as follows. The first section 102 includes a thermally insulating first layer 108 (FIG. 3), or inner layer, that is disposed about the first exhaust conduit 16. In some embodiments, the thermally insulating first layer 108 is a ceramic fiber fabric or blanket. In certain embodiments, the thermally insulating first layer 108 includes one or more of the following properties: fire-proof at a constant temperature of 2,400 degrees Fahrenheit; flexible to facilitate positioning about the first exhaust conduit 16; light weight; low heat transfer; sufficient for use in high-temperature applications, such as in kilns, pizza ovens, furnaces, and forges. In certain embodiments, the thermally insulating first layer 108 has a thickness of about 1 inch. In some embodiments, the thermally insulating first layer 108 has an area density of about 542 g per square meter. In some embodiments, the first layer 108 is coupled to the first exhaust conduit 16 via one or more fasteners (shown elsewhere), more specifically one or more non-reversibly releasable fasteners (as used herein, non-reversibly releasable fasteners refer to fasteners that are only releasable by damaging or destroying the fasteners and therefore are not reusable), such as cable ties, more specifically metallic cable ties, and even more specifically stainless steel cable ties.

[0023] The first section 102 of the catalytic converter anti-theft system 100 further includes one or more cables, illustratively a first cable 110 and a second cable 112, disposed about the thermally insulating first layer 108 and thereby the first exhaust conduit 16. In some embodiments, the first cable 110 and the second cable 112 extend in a

figure-eight configuration about the thermally insulating first layer 108; stated another way, the first cable 110 extends in a first helical direction about the thermally insulating first layer 108 and the second cable 112 extends in a second, opposite helical direction about the thermally insulating first layer 108. In certain embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, are coupled to the frame 30 of the vehicle 10 via one or more weldments 114, such as tack weldments formed in a metal inert gas (“MIG”) welding process. In some embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, are metallic cables, more specifically galvanized cables. In some embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, have a diameter of about $\frac{3}{16}$ inch. In some embodiments and with additional reference to FIG. 4, one or more of the cables, for example the first cable 110 and/or the second cable 112, have a 7×19 construction (that is, the cable includes 7 strands 116 each having 19 wires 118, the cable thereby having a total of 133 wires 118). In certain embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, include one or more of the following properties: flexible to facilitate positioning about the thermally insulating first layer 108; corrosion-resistant; cut-resistant. In some embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, are secured or further secured to the exhaust system 12 via one or more fasteners (shown elsewhere), more specifically one or more non-reversibly releasable fasteners, such as cable ties, more specifically metallic cable ties, and even more specifically stainless steel cable ties.

[0024] Referring again to FIG. 3, the first section 102 of the catalytic converter anti-theft system 100 further includes a cut and abrasion resistant second layer 120, or intermediate layer, that is disposed about the one or more cable(s), for example the first cable 110 and the second cable 112, and thereby the thermally insulating first layer 108 and the first exhaust conduit 16. In some embodiments, the cut and abrasion resistant second layer 120 is an aramid fiber layer, more specifically a para-aramid fiber layer, for example a Kevlar® layer, a meta-aramid fiber layer, for example a Nomex® layer, or a layer including both para-aramid fibers and meta-aramid fibers. In some embodiments, the second layer 120 is formed as an interwoven twill weave. In certain embodiments, the cut and abrasion resistant second layer 120 includes one or more of the following properties: fire-retardant up to 700 degrees Fahrenheit; water repellent; flexible; lightweight; sufficient for use in protective clothing, such as chainsaw pants, gloves, firefighter suits, and body armor. In some embodiments, the cut and abrasion resistant second layer 120 is a 50% Nomex® and a 50% Kevlar® cross weave twill pattern. In certain embodiments, the cut and abrasion resistant second layer 120 has an area density of about 237 g per square meter. In some embodiments, the second layer 120 is secured or further secured to the exhaust system 12 via one or more fasteners (shown elsewhere), more specifically one or more reversibly releasable fasteners (as used herein, reversibly releasable fasteners refer to fasteners that are releasable without damaging or destroying the fasteners, thereby permitting their reuse), such as worm drive clamps, more specifically metallic worm drive clamps, and even more specifically galvanized worm drive clamps.

[0025] With further reference to FIGS. 2 and 3, the first section 102 of the catalytic converter anti-theft system 100 further includes an abrasion resistant third layer 122 disposed about the cut and abrasion resistant second layer 120, and thereby the cable(s), for example the first cable 110 and/or the second cable 112, the thermally insulating first layer 108, and the first exhaust conduit 16. In some embodiments, the abrasion resistant third layer 122 is a carbon fiber layer. In some embodiments, the abrasion resistant third layer 122 is formed as an interwoven twill weave, more specifically a 2x2 twill weave. In certain embodiments, the abrasion resistant third layer 122 includes one or more of the following properties: water repellant; flexible; lightweight; high tensile strength; high durability; sufficient for use as a structural reinforcement. In certain embodiments, the abrasion resistant third layer 122 has an area density of about 220 g per square meter. In some embodiments, the abrasion resistant third layer 122 has a thickness of 0.4 mm to 0.5 mm, more specifically about 0.46 mm. In some embodiments, the third layer 122 is secured or further secured to the exhaust system 12 via one or more fasteners 124 (FIG. 2), more specifically one or more reversibly releasable fasteners, such as worm drive clamps, more specifically metallic worm drive clamps, and even more specifically galvanized worm drive clamps. In certain embodiments, the third layer 122 is coupled to the second layer 120 in alternative or additional manners, for example fusing or stitching together the edges of the layers.

[0026] With reference again to FIG. 2, the second section 104 of the catalytic converter anti-theft system 100 includes one or more cables, more specifically the first cable 110 and the second cable 112 extending continuously from the first section 102 of the system 100. In certain embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, are disposed about the catalytic converter 18. In some embodiments, the first cable 110 and the second cable 112 extend in a figure-eight configuration about the catalytic converter 18; stated another way, the first cable 110 extends in a first helical direction about the catalytic converter 18 and the second cable 112 extends in a second, opposite helical direction about the catalytic converter 18. In certain embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, are coupled to the catalytic converter 18 via one or more weldments 126, such as tack weldments formed in a MIG welding process.

[0027] With reference to FIGS. 2 and 5, the third section 106 of the catalytic converter anti-theft system 100 is as follows. The third section 106 includes a thermally insulating first layer 128 (FIG. 5), or inner layer, that is disposed about the second exhaust conduit 20. In some embodiments, the thermally insulating first layer 128 is the same or substantially similar to the thermally insulating first layer 108 of the first section 102 of the system 110 (that is, having the same or substantially similar features, properties, and/or characteristics as the thermally insulating first layer 108). In some embodiments, the first layer 128 is coupled to the second exhaust conduit 20 via one or more fasteners (shown elsewhere), more specifically one or more non-reversibly releasable fasteners, such as cable ties, more specifically metallic cable ties, and even more specifically stainless steel cable ties.

[0028] The third section 106 also includes one or more cables, more specifically the first cable 110 and the second

cable 112 extending continuously from the second section 104 of the system 100. In certain embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, are disposed about the thermally insulating first layer 128 and thereby the second exhaust conduit 20. In some embodiments, the first cable 110 and the second cable 112 extend in a figure-eight configuration about the thermally insulating first layer 128; stated another way, the first cable 110 extends in a first helical direction about the thermally insulating first layer 128 and the second cable 112 extends in a second, opposite helical direction about the thermally insulating first layer 128. In certain embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, are coupled to the frame 30 of the vehicle 10 via one or more weldments 130 (FIG. 2), such as tack weldments formed in a MIG welding process. In some embodiments, one or more of the cables, for example the first cable 110 and/or the second cable 112, are secured or further secured to the exhaust system 12 via one or more fasteners (shown elsewhere), more specifically one or more non-reversibly releasable fasteners, such as cable ties, more specifically metallic cable ties, and even more specifically stainless steel cable ties.

[0029] The third section 106 of the catalytic converter anti-theft system 100 further includes a cut and abrasion resistant second layer 132 (FIG. 5), or intermediate layer, that is disposed about the one or more cable(s), for example the first cable 110 and the second cable 112, and thereby the thermally insulating first layer 128 and the second exhaust conduit 20. In some embodiments, the cut and abrasion resistant second layer 132 is the same or substantially similar to the cut and abrasion resistant second layer 120 of the first section 102 of the system 110 (that is, having the same or substantially similar features, properties, and/or characteristics as the cut and abrasion resistant second layer 120). In some embodiments, the second layer 132 is secured or further secured to the exhaust system 12 via one or more fasteners (shown elsewhere), more specifically one or more reversibly releasable fasteners (as used herein, reversibly releasable fasteners refer to fasteners that are releasable without damaging or destroying the fasteners, thereby permitting their reuse), such as worm drive clamps, more specifically metallic worm drive clamps, and even more specifically galvanized worm drive clamps.

[0030] The third section 106 of the catalytic converter anti-theft system 100 additionally includes an abrasion resistant third layer 134 disposed about the cut and abrasion resistant second layer 132, and thereby the cable(s), for example the first cable 110 and/or the second cable 112, the thermally insulating first layer 128, and the second exhaust conduit 20. In some embodiments, the abrasion resistant third layer 134 is the same or substantially similar to the abrasion resistant third layer 122 of the first section 102 of the system 110 (that is, having the same or substantially similar features, properties, and/or characteristics as the abrasion resistant third layer 122). In some embodiments, the third layer 122 is secured or further secured to the exhaust system 12 via one or more fasteners 136 (FIG. 2), more specifically one or more reversibly releasable fasteners, such as worm drive clamps, more specifically metallic worm drive clamps, and even more specifically galvanized worm drive clamps.

[0031] In certain embodiments, the catalytic converter anti-theft system 100 facilitates maintaining the catalytic

converter **18** at an appropriate working temperature. More specifically and as illustrated, one or more of the layers of the first section **102** of the system **100** (for example, the thermally insulating first layer **108**, the cut and abrasion resistant second layer **120**, and/or the abrasion resistant third layer **122**) are disposed apart from, or are not disposed about, the catalytic converter **18**, and one or more of the layers of the third section **106** of the system **100** (for example, the thermally insulating first layer **128**, the cut and abrasion resistant second layer **132**, and/or the abrasion resistant third layer **134**) are disposed apart from, or are not disposed about, the catalytic converter **18**.

[0032] FIG. 6 shows a method **200** for coupling the catalytic converter anti-theft system **100** to an exhaust system of a vehicle, according to an example embodiment, and FIGS. 7-9 show actions associated with the method **200**. The following description of the method **200** describes coupling components of the first section **102** of the system **100** to the vehicle, although similar actions may be used, simultaneously or non-simultaneously, to couple components of the third section **106** of the system **100** to the vehicle. The method begins at block **202**, and as shown in FIG. 7, by positioning the thermally insulating first layer **108** about at least a portion of the exhaust system, for example the first exhaust conduit **16**. In some embodiments, the thermally insulating first layer **108** is disposed apart from the catalytic converter (shown elsewhere). At block **204** and as shown in FIG. 7, one or more fasteners **138**, such as cable ties, are positioned about the thermally insulating first layer **108** and thereby the first exhaust conduit **16**. At block **206**, one or more cables, for example the first cable **110** and the second cable **112** (shown elsewhere), are positioned about the fasteners **138**, and thereby the thermally insulating first layer **108** and the first exhaust conduit **16**. In certain embodiments, the cables are also positioned about the catalytic converter. In some embodiments, the cables are positioned in a figure-eight configuration about the first exhaust conduit **16** and/or the catalytic converter. At block **208**, the cables, for example the first cable **110** and the second cable **112**, are coupled to the vehicle (shown elsewhere) and/or the catalytic converter, more specifically the cables are welded to the frame of the vehicle and/or the catalytic converter. At block **210** and as shown in FIG. 8, the method **200** continues by positioning the cut and abrasion resistant second layer **120** about the cables and thereby the fasteners **138**, the thermally insulating first layer **108**, and the first exhaust conduit **16**. In some embodiments, the cut and abrasion resistant second layer **120** is disposed apart from the catalytic converter. At block **212** and as shown in FIG. 8, one or more fasteners **140**, such as worm drive clamps, are positioned about the cut and abrasion resistant second layer **120** and thereby the cables, the fasteners **138**, the thermally insulating first layer **108**, and the first exhaust conduit **16**. At block **214** and as shown in FIG. 9, the method **200** continues by positioning the abrasion resistant third layer **122** about the fasteners **140** and thereby the cut and abrasion resistant second layer **120**, the cables, the fasteners **138**, the thermally insulating first layer **108**, and the first exhaust conduit **16**. In some embodiments, the abrasion resistant third layer **122** is disposed apart from the catalytic converter. At block **216** and as shown in FIG. 9, one or more of the fasteners **124** are positioned about the abrasion resistant third layer **122** and thereby the fasteners **140**, the cut and abrasion resistant second layer **120**,

the cables, the fasteners **138**, the thermally insulating first layer **108**, and the first exhaust conduit **16**.

[0033] The description and illustration of one or more aspects provided in this application are not intended to limit or restrict the scope of the following claims in any way. The aspects, examples, and details provided in this application are considered sufficient to convey possession and enable others to make and use embodiments according to the present disclosure. The following claims should not be construed as being limited to any aspect, example, or detail provided in this application. Regardless of whether shown and described in combination or separately, the various features (both structural and methodological) are intended to be selectively included or omitted to produce an embodiment with a particular set of features. Having been provided with the description and illustration of exemplary embodiments, one skilled in the art may envision variations, modifications, and alternate aspects falling within the spirit of the broader aspects of the disclosure of this application that do not depart from the scope of the following claims.

[0034] The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the disclosed embodiments and does not pose a limitation on the scope of the disclosed embodiments unless otherwise claimed. Numerous modifications and adaptations will be readily apparent to those of ordinary skill in this art.

What is claimed is:

1. A catalytic converter anti-theft system for an exhaust system of a vehicle, the system comprising:
 - a thermally insulating first layer configured to be disposed about at least a portion of the exhaust system;
 - a cable configured to be disposed about the thermally insulating first layer;
 - a cut and abrasion resistant second layer configured to be disposed about the cable; and
 - an abrasion resistant third layer configured to be disposed about the cut and abrasion resistant second layer.
2. The anti-theft system of claim 1, wherein the thermally insulating first layer is a ceramic fiber blanket.
3. The anti-theft system of claim 1, wherein the cable is a galvanized cable.
4. The anti-theft system of claim 1, wherein the cut and abrasion resistant second layer is an aramid fiber layer.
5. The anti-theft system of claim 1, wherein the abrasion resistant third layer is a carbon fiber layer.
6. The anti-theft system of claim 1, further comprising a plurality of fasteners configured to couple the anti-theft system to the vehicle.
7. The anti-theft system of claim 6, wherein the plurality of fasteners comprise worm drive clamps disposed about the abrasion resistant third layer.
8. The anti-theft system of claim 6, wherein the plurality of fasteners is a first plurality of fasteners, the first plurality of fasteners being disposed about the abrasion resistant third layer, and further comprising a second plurality of fasteners, the second plurality of fasteners being disposed about the thermally insulating first layer.
9. The anti-theft system of claim 8, wherein the second plurality of fasteners comprise cable ties.
10. The anti-theft system of claim 1, wherein the thermally insulating first layer is a first thermally insulating first layer, the cut and abrasion resistant second layer is a first cut and abrasion resistant second layer, and the abrasion resis-

tant third layer is a first abrasion resistant third layer, and wherein the anti-theft system comprises:

- a first section comprising the first thermally insulating first layer, the first cut and abrasion resistant second layer, the cable, and the first abrasion resistant third layer;
- a second section comprising the cable, the cable extending continuously from the first section and configured to be disposed about at least a portion of the exhaust system;
- a third section comprising:
 - a second thermally insulating first layer configured to be disposed about at least a portion of the exhaust system;
 - the cable, the cable extending continuously from the second section and configured to be disposed about the second thermally insulating first layer;
 - a second cut and abrasion resistant second layer configured to be disposed about the cable; and
 - a second abrasion resistant third layer configured to be disposed about the second cut and abrasion resistant second layer.

11. The anti-theft system of claim **10**, wherein the cable is a first cable, and further comprising a second cable configured to be disposed about the first thermally insulating first layer in the first section, the second cable extending continuously from the first section to the second section and configured to be disposed about at least a portion of the exhaust system in the second section, the second cable extending continuously from the second section to the third section and configured to be disposed about the second thermally insulating first layer in the third section.

12. A catalytic converter anti-theft system for an exhaust system of a vehicle, the system comprising:

- a thermally insulating first layer configured to be disposed about at least a portion of the exhaust system;
- a cut and abrasion resistant second layer configured to be disposed about the thermally insulating first layer;
- an abrasion resistant third layer configured to be disposed about the cut and abrasion resistant second layer; and
- a plurality of fasteners configured to couple the anti-theft system to the vehicle.

13. The anti-theft system of claim **12**, wherein the thermally insulating first layer is a ceramic fiber blanket.

14. The anti-theft system of claim **12**, wherein the cut and abrasion resistant second layer is an aramid fiber layer.

15. The anti-theft system of claim **12**, wherein the abrasion resistant third layer is a carbon fiber layer.

16. The anti-theft system of claim **12**, wherein the plurality of fasteners comprise worm drive clamps disposed about the abrasion resistant third layer.

17. The anti-theft system of claim **12**, wherein the plurality of fasteners is a first plurality of fasteners, the first plurality of fasteners being disposed about the abrasion resistant third layer, and further comprising a second plurality of fasteners, the second plurality of fasteners being disposed about the thermally insulating first layer.

18. The anti-theft system of claim **17**, wherein the second plurality of fasteners comprise cable ties.

19. A method for coupling a catalytic converter anti-theft system to an exhaust system of a vehicle, the method comprising:

- positioning a thermally insulating first layer about at least a portion of the exhaust system;
- positioning a cable about the thermally insulating first layer;
- coupling the cable to the vehicle;
- positioning a cut and abrasion resistant second layer about the cable;
- positioning an abrasion resistant third layer about the cut and abrasion resistant second layer; and
- coupling the abrasion resistant third layer to the vehicle using a plurality of fasteners.

20. The method of claim **19**, wherein the exhaust system comprises an exhaust conduit and a catalytic converter coupled to the exhaust conduit, and the thermally insulating first layer, the cut and abrasion resistant second layer, and the abrasion resistant third layer are positioned about the exhaust conduit and disposed apart from the catalytic converter.

21. The method of claim **20**, wherein coupling the cable to the vehicle comprises welding the cable to the catalytic converter.

22. The method of claim **20**, wherein coupling the cable to the vehicle comprises welding the cable to a frame of the vehicle.

23. The method of claim **19**, wherein the cable is a first cable, and positioning the first cable about the thermally insulating first layer comprises positioning the first cable and a second cable about the thermally insulating first layer and a catalytic converter of the exhaust system.

24. The method of claim **23**, wherein positioning the first cable and the second cable about the thermally insulating first layer and the catalytic converter comprises positioning the first cable and the second cable in a figure-eight configuration about the thermally insulating first layer and the catalytic converter.

25. The method of claim **19**, wherein the plurality of fasteners comprise worm drive clamps, and coupling the abrasion resistant third layer to the vehicle comprises positioning the worm drive clamps about the abrasion resistant third layer.

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